Database ERD and EERD Visual Components Reference

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October 14, 2026

1 Introduction

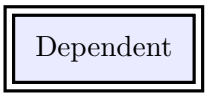
The ERD and EERD symbols from the ITI Database course, in Chen notation (Elmasri & Navathe's version). The TikZ source for every figure is in `erd_components.tex` and `erdstyles.sty`.

2 Core ERD Components

2.1 Entity Types

An entity type is a person, place, object, or concept the organization needs to store data about.

Strong Entity (Independent)



Weak Entity (Dependent)

2.2 Attribute Types

Attributes are properties of entities or relationships.

Simple Attribute

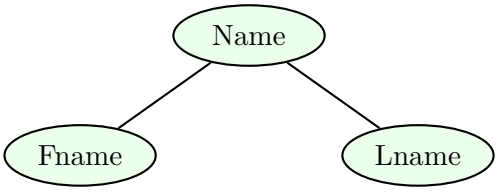
Key Attribute

Partial Key (weak entity)

Derived Attribute

Multivalued Attribute

2.3 Advanced Attributes

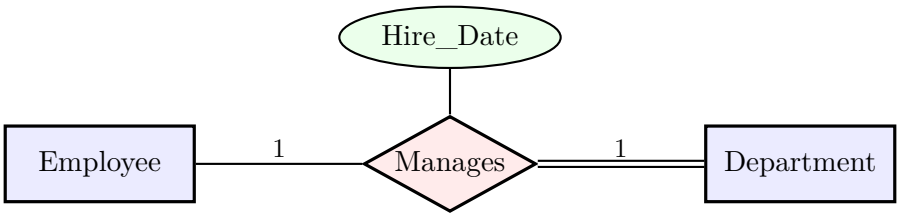


Composite Attribute

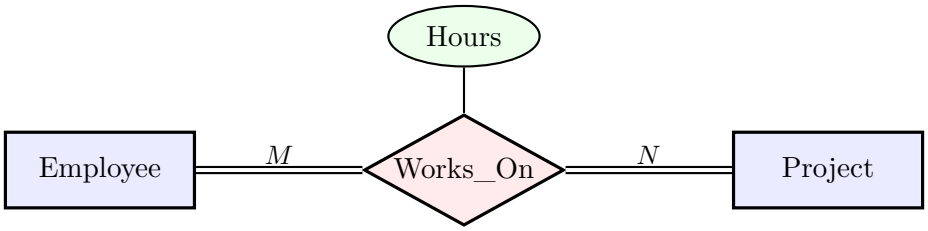
Complex Attribute (Multivalued + Composite)

2.4 Relationships and Participation Constraints

Relationships represent associations between entities. The cardinality ratio (1:1, 1:M, M:N) is written on the lines. Participation states whether every instance must take part: a **single line** means partial (optional) participation, a **double line** means total (mandatory) participation.



Binary One-to-One (1:1) relationship with an attribute. Single line: participation of Employee is partial (not every employee is a manager). Double line: participation of Department is total (every department must have a manager).

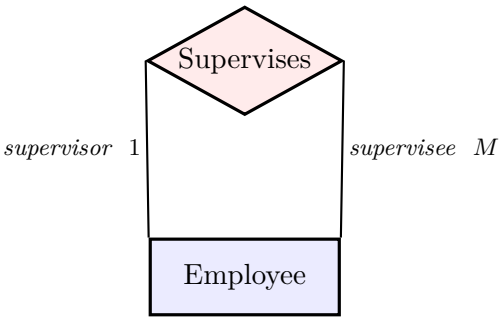


Binary Many-to-Many (M:N) relationship with total participation on both sides.

Identifying Relationship (double diamond) connecting the owner (strong) entity to a weak entity. The weak entity always has total participation.

2.5 Unary (Recursive) Relationships

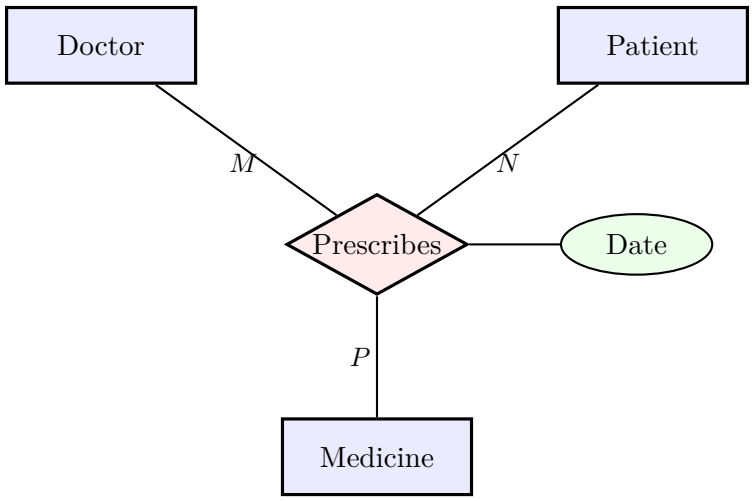
Unary relationships occur between instances of the same entity type. Role names distinguish the two ends.



Unary 1:M Relationship (Supervision)

2.6 Ternary Relationships

Ternary relationships connect three distinct entity types.

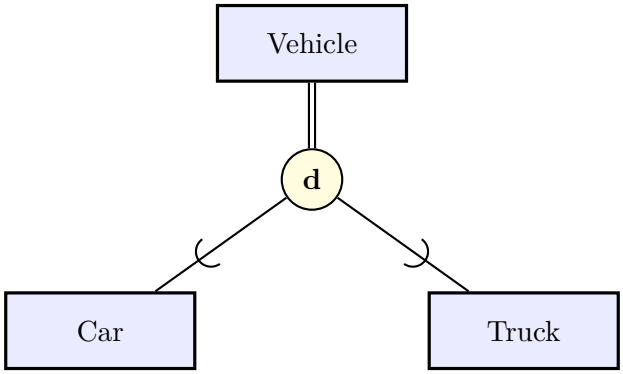


Ternary M:N:P Relationship with a relationship attribute (Date).

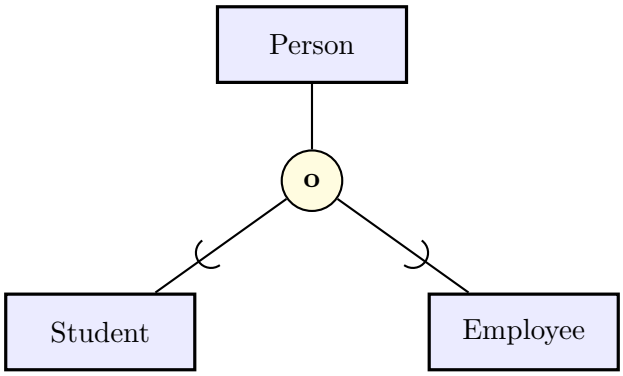
3 Enhanced ERD (EERD) Components

EERD models supertypes, subtypes, and specialization/generalization hierarchies. The line from the supertype to the circle shows *completeness* (double = total, single = partial). The letter in the circle shows *disjointness* (d = disjoint, o = overlapping). The U-shaped subset symbol on each subtype line opens toward the circle and means "subtype $\subset$ supertype".

3.1 Specialization Hierarchy with Completeness and Disjoint Constraints



Total Disjoint Specialization: Double line to circle = Total Specialization (every Vehicle must be either a Car or a Truck). 'd' in circle = Disjoint (a Vehicle cannot be both a Car and a Truck).



Partial Overlapping Specialization: Single line to circle = Partial Specialization (a Person can exist without being a Student or an Employee). 'o' in circle = Overlap (a Person can be both a Student and an Employee simultaneously).

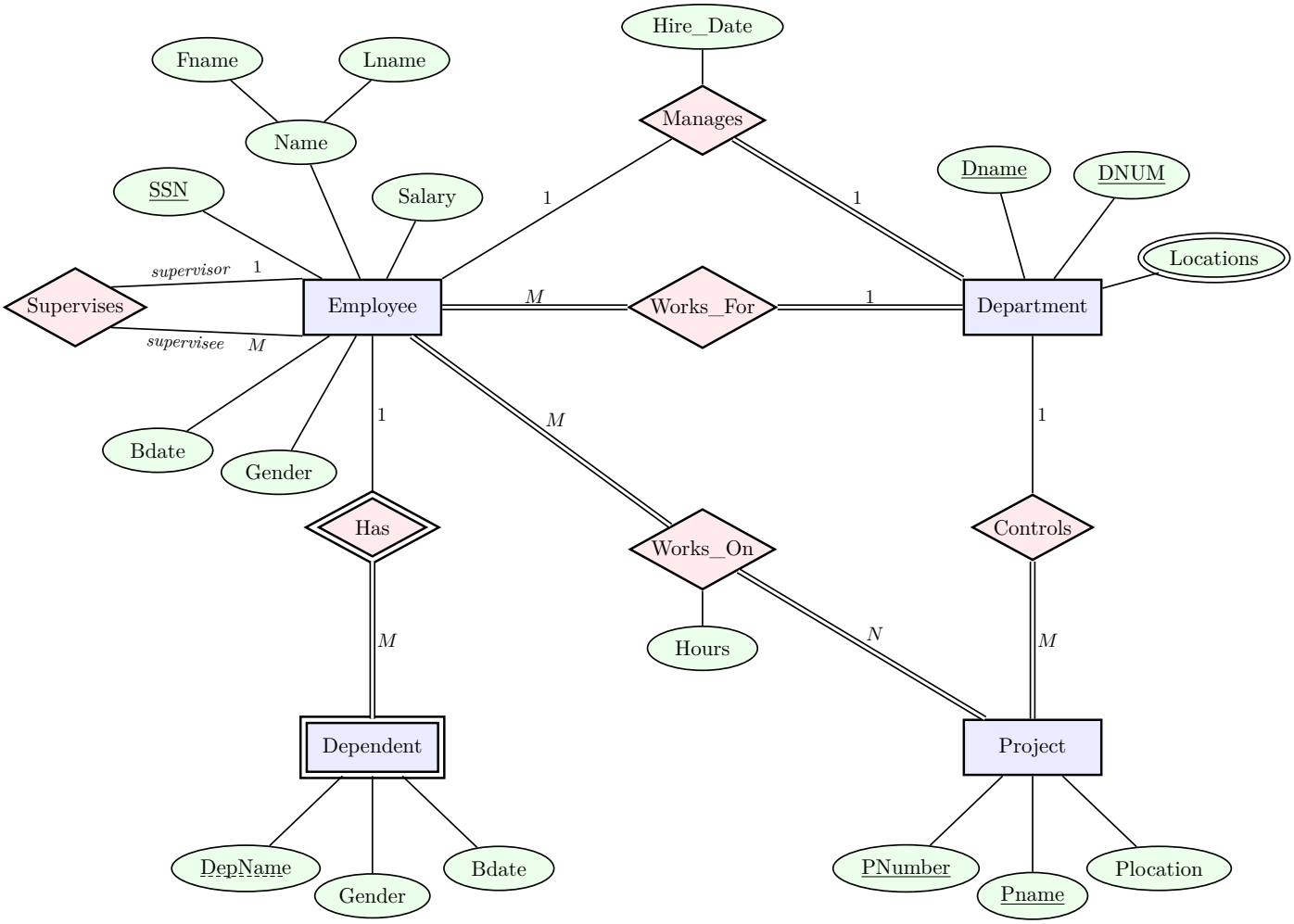
4 Complete Case Studies

4.1 Company Case Study ERD

From the description on page 4 of the lecture notes:

- Employees: SSN (key), Bdate, Gender, Salary, Name (Fname, Lname).
- Departments: DNUM (key), Dname (key, unique), Locations (multivalued).
- Projects: PNumber (key), Pname (key, unique), Plocation.
- Dependents (weak entity): DepName (partial key), Gender, Bdate.

Relationship	**Cardinality**	**Participation**
Works_For	Employee M : 1 Department	Both total
Manages	Employee 1 : 1 Department	Employee partial, Department total; attribute `Hire_Date`
Controls	Department 1 : M Project	Department partial, Project total
Works_On	Employee M : N Project	Both total; attribute `Hours`
Has	Employee 1 : M Dependent	Employee partial, Dependent total (identifying)
Supervises	Employee 1 : M Employee	Partial on both roles (recursive)

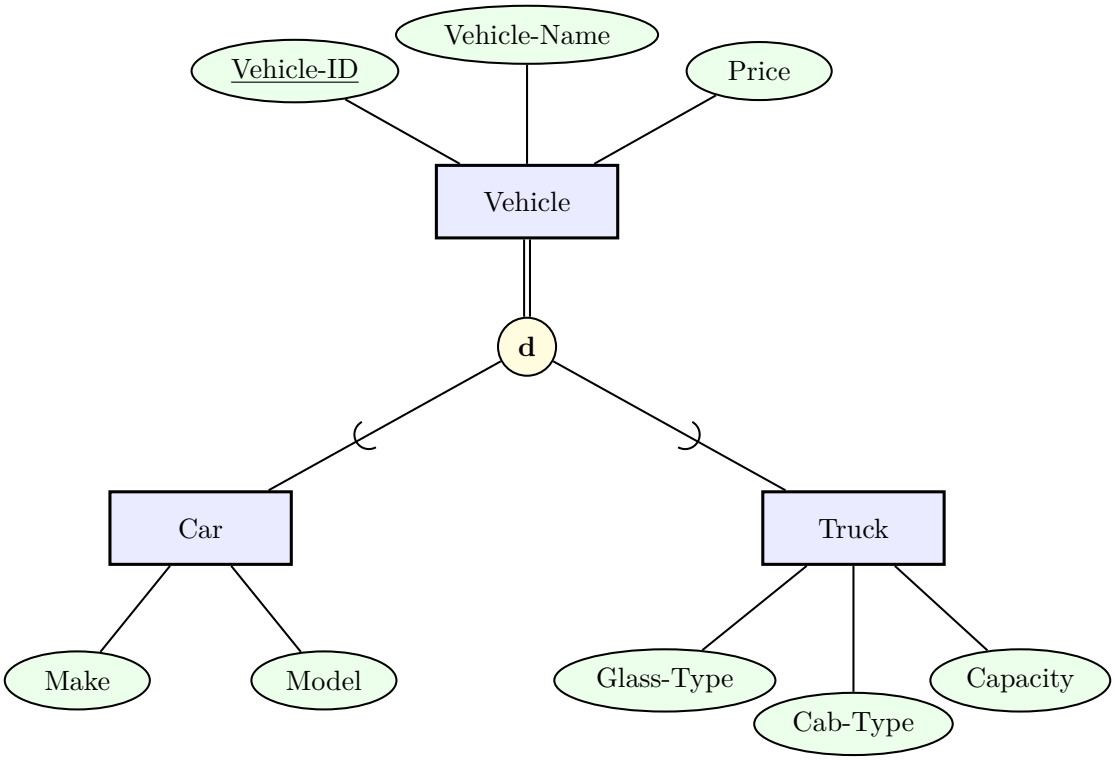


Company Database Entity-Relationship Diagram

4.2 Vehicle Specialization Case Study EERD

From page 5/11 of the lecture notes:

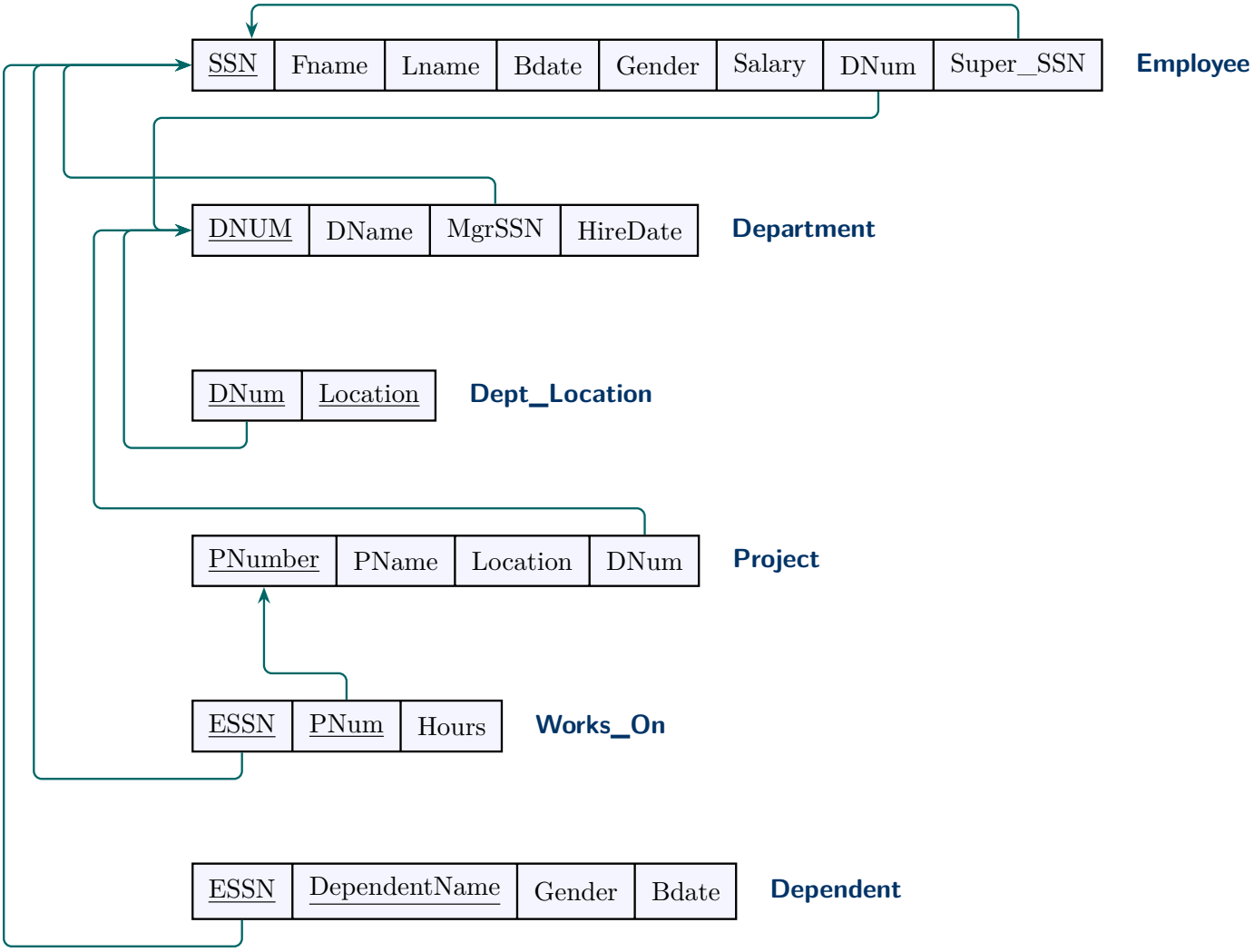
- Supertype: Vehicle with attributes Vehicle-ID (key), Vehicle-Name, Price.
- Specialization circle with disjoint (d) and total specialization (double line).
- Subtypes:
  - Car with attributes Make, Model.
  - Truck with attributes Glass-Type, Cab-Type, Capacity.



Vehicle Specialization Enhanced Entity-Relationship Diagram

5 Relational Schema Diagram

After mapping, the Company ERD becomes the following tables. Primary keys are underlined, and each arrow runs from a foreign key to the primary key it references. Note the arrow from `Super_SSN` back into its own table (the recursive Supervises relationship) and the pair of arrows between `Employee` and `Department`, which form a circular reference.



Relational schema of the Company database